Quiz C

volume solid generated region bounded by $y=x^2$ and $y=0$ $[0,2]$ ab y axis

(1)

1. $2\pi\int_0^2 x(x^2)\,dx = V$

$$2\pi\left(\frac{x^4}{4}\right)\Big|_0^2 = 2\pi\left(\frac{16}{4}-0\right)$$

$$= 2\pi(4) = \boxed{8\pi}$$

volume of solid generated revolving region bounded by $y=x$, $y=0$ interval $[0,1]$ ab line $x=2$

2. $V = 2\pi\int_0^1 (2-x)(x)\,dx$

$$= 2\pi\int_0^1 (2x-x^2)\,dx$$

$$= 2\pi\left(x^2-\frac{x^3}{3}\right)\Big|_0^1$$

$$= 2\pi\left(1-\frac{1}{3}\right)-0 = \frac{2}{3}\cdot 2\pi = \boxed{\frac{4\pi}{3}}$$

# Quiz D

$$L=\int_a^b \sqrt{1+f'(x)^2}\,dx$$

$x^3$ $3x^2$ $x^{-1}$ $-1x^{-2}$

$\frac{1}{2}(x^{-1})$ $-\frac{1}{2}x^{-2}$

a. $y=\frac{1}{6}x^3+\frac{1}{2}x^{-1}$ $[1,2]$

$$\frac{dy}{dx}=\frac{1}{2}x^2-\frac{1}{2x^2}=\frac{x^2}{2}-\frac{1}{2x^2}$$

$$\left(\frac{dy}{dx}\right)^2=\left(\frac{x^2}{2}-\frac{1}{2x^2}\right)\left(\frac{x^2}{2}-\frac{1}{2x^2}\right)=\frac{1}{4}x^4-\frac{1}{4}-\frac{1}{4}+\frac{1}{4x^4}$$

$$\left(\frac{dy}{dx}\right)^2+1=\frac{1}{4}x^4-\frac{1}{2}+\frac{1}{4x^4}+1=\frac{x^4}{4}+\frac{1}{2}+\frac{1}{4x^4}$$

$$L=\int_1^2\sqrt{1+\left(\frac{x^2}{2}-\frac{1}{2x^2}\right)^2}\,dx$$

$\left(\frac{x^2}{2}+\frac{1}{\sqrt{2}}\right)$ $\frac{x^2}{2}+\frac{1}{\sqrt{2}}$

$$L=\int_1^2\sqrt{\frac{x^4}{4}+\frac{1}{2}+\frac{1}{4x^4}}$$

$\frac{2x^2}{4x^2}$ $x^2$ $\frac{1}{3}x^3$

$$L=\int_1^2\left(\frac{x^2}{2}+\sqrt{\frac{1}{2}}+\frac{1}{2x^2}\right)dx$$

$-\frac{1}{2}\cdot-\frac{1}{2}$ $=-1+1$ $\frac{1}{6}x^3$

$$L=\frac{x^3}{6}+\sqrt{\frac{1}{2}}x-\frac{1}{2x}\Big|_1^2$$

$\frac{1}{2}x^{-2}$

$$=\left(\frac{8}{6}+\sqrt{\frac{1}{2}}-\frac{1}{4}\right)-\left(\frac{1}{6}+\frac{1}{\sqrt{2}}-\frac{1}{2}\right)$$

$$\frac{8}{6}+\cancel{\sqrt{\frac{1}{2}}}-\frac{1}{4}-\frac{1}{6}\cancel{-\sqrt{\frac{1}{2}}}+\frac{1}{2}$$

$$=\frac{7}{6}-\frac{1}{4}+\frac{2}{4}$$

$$=\frac{7}{6}+\frac{1}{4}$$

$$=\frac{14}{12}+\frac{3}{12}=\frac{17}{12}$$

$x = \frac{1}{3}y^3 + \frac{1}{4y}$ , $[1, 2]$ $\quad \frac{y^3}{3y^2} \quad \frac{y^{-1}}{-1y^{-2}}$

$g'(y) = y^2 - \frac{1}{4y^2}\,dy$

$(g'(y))^2 = \left(y^2 - \frac{1}{4y^2}\right)\left(y^2 - \frac{1}{4y^2}\right)$

$= y^4 - \frac{y^2}{4y^2} - \frac{y^2}{4y^2} + \frac{1}{16y^4} = y^4 - \frac{1}{2y^2} + \frac{1}{16y^4}$

$L = \int_1^2 \sqrt{1 + (g'(y))^2}\,dy$

$= \int_1^2 \sqrt{1 + y^4 - \frac{1}{2y^2} + \frac{1}{16y^4}}\,dy$ $\quad y^{-2}$

$= \int_1^2 1 + y^2 - \sqrt{\frac{1}{2y^2}} + \frac{1}{4y^2}\,dy$

$= y + \frac{y^3}{3} - \int \sqrt{\frac{1}{2y^2}} - \frac{1}{4y} \Big|_1^2$